

Causation

PHIL UA-78 Metaphysics

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Introduction to causation

Ways of talking about causation

- ▶ The lighting of the fuse caused the explosion.
- ▶ The lighting of the fuse caused the bomb to explode.
- ▶ The lighting of the fuse was a cause of the bomb's exploding.
- ▶ The bomb exploded because the fuse was lit.

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NB: we say 'a cause', not 'the cause'. Uncontroversially, most events have *many* causes. The question what's going on when we single out one as 'the' cause is interesting, but it plausibly introduces a whole extra layer of messy context-sensitivity.

Some background on the concept of an event, which hasn't been a big player in the course so far.

1. To any event e there corresponds a state of affairs—the state of affairs that e occurs, $O(e)$. But perhaps not every state of affairs p is such that there is an event e for which p is equivalent to $O(e)$. (Example: the proposition that either Obama is president of the US in 2010 or Gaddafi is president of Libya in 2008.)

2. Do not take it for granted that when we refer to an event using a name of the form 'the ϕ -ing of x ', e.g. 'the death of Socrates', then necessarily, that event occurs iff x ϕ s at some time or other. Perhaps the particular event that was the actual death of Socrates would not have occurred if Socrates had died peacefully in his sleep five years later.

3. Some events stand in "constitutive" relations. For example: the Battle of Britain partly constituted World War II; Socrates' death partly constituted Xanthippe's widowing. This kind of intimate connection seems different from mere causation. To put a label on it: say that two events are "disjoint" when they are not related in this way.

Nomic sufficiency analyses of causation

The simple nomic sufficiency analysis

There's a widespread sense that causation is, somehow, closely related to laws of nature. Here's a simple idea about how they are related:

Simple nomic sufficiency analysis of causation

C is a cause of E iff C and E are disjoint and C is a *nomically sufficient condition* for E —i.e., iff it is nomically necessary that if C occurs, E occurs.

This proposal faces many problems. We'll discuss five of them.

1. The problem of effects

In some cases, A is nomically sufficient for B although, intuitively, B is a cause of A and not vice versa.

For example: the presence at $t + \Delta$ of an alpha-particle shooting out from a nucleus might be a nomically sufficient condition for the decay of the nucleus at t .

2. The problem of epiphenomena

In some cases, A is nomically sufficient for B although intuitively, neither is a cause of the other, but rather they are effects of a common cause.

For example: the presence at t of an alpha-particle with such and such energy shooting out from a nucleus might be a nomically sufficient condition for the presence at t , or at some other earlier or later time, of a beta-particle with such and such energy shooting out on the opposite side.

3. The problem of “completeness”

Under realistic physical assumptions, the only facts about a time t that nomically necessitate the occurrence of some ordinary event E at a later time t' are very rich facts that characterise the state of some very wide spatial region at t .

(For example: in relativity theory, the intersection of the entire past light cone of E with t .)

4. The problem of pre-emption

Two trails of gunpowder are laid out on the ground, both leading to a pile of TNT. Wile E. Coyote lights the long trail; Road Runner lights the short one. The bomb explodes when the fire reaches it along the short trail.

The lighting of the long trail—in conjunction with lots of “background” facts—is nomically sufficient for the explosion. But it is not (even taken together with all those background facts) a cause of the explosion.

5. The problem of chancy causation

Suppose the world is “thoroughly chancy” in the following sense: for any t , any nomically possible state of affairs entirely about how history goes after t has some chance of being realised at t .

Then, no fact entirely about an earlier time ever nomically necessitates a fact entirely about a later time, unless the latter fact is nomically necessary all by itself.

Given the simple nomic sufficiency analysis, it follows that no earlier event ever causes a later one!

Refinements of the nomic sufficiency analysis

The main focus has been getting around problem 3, the problem of “completeness”. A natural idea here is that a cause shouldn’t have to be nomically sufficient for the effect all by itself, but only needs to be sufficient in combination with certain ‘background conditions’. This might suggest a revision that merely requires causes to be *parts* of nomically sufficient conditions:

Partial sufficiency analysis (broken)

C is a cause of E iff C and E are disjoint and there are some other events $D_1, \dots, D_n$, disjoint from both C and E , such that it’s nomically necessary that if C and $D_1, \dots, D_n$ all occur, E occurs.

But this is no good at all: if there exist any $D_1, \dots, D_n$ which are disjoint from E and nomically necessitate E , then *every* event C that is disjoint from E and from all of $D_1, \dots, D_n$ will count as a cause of E !

Refinements of the nomic sufficiency analysis

Refined partial sufficiency analysis

C is a cause of E iff C and E are disjoint and for some other events $D_1, \dots, D_n$, disjoint from C and E : the occurrence of C together with $D_1, \dots, D_n$ is nomically sufficient for the occurrence of E , but the occurrence of $D_1, \dots, D_n$ is not nomically sufficient for the occurrence of E .

For short: C is part of a minimal nomically sufficient condition for E .

- Mackie calls this an 'INUS' condition: an 'insufficient but necessary part of an unnecessary but sufficient condition'.

This proposal would be just as hopeless as the previous one if we allowed for “disjunctive” events like *Either Booth pulls the trigger and this tree falls in Siberia, or Booth does not pull the trigger and the tree does not fall.*

Well, maybe we need a theory of events that rules out events like that.

But anyway the other four problems remain unaddressed!

The counterfactual analysis

The simple counterfactual analysis

These problems caused Lewis to propose a different idea, versions of which have been very influential since:

Simple counterfactual analysis

C is a cause of E iff C and E are disjoint and E *counterfactually depends* on C —i.e., if C hadn't occurred, E wouldn't have occurred.

Remember that counterfactuals are context-sensitive. And even though 'is a cause of' is also plausibly context-sensitive, the analysis does not look promising when the counterfactual is read in the “backtracking” way that's natural when we say things like 'If this happened, that would have to have been because that happened.'

But, let's assume we can set those readings aside and zero in on a class of 'standard' readings. Then the analysis looks like it can help with some of our problems.

1. Problem of effects

2. Problem of epiphenomena

- ▶ So long as we can understand the relevant counterfactuals in such a way that there is not widespread counterfactual dependence of later events on earlier ones, the counterfactual analysis does much better with these two problems.
- ▶ There is still a worry about short-range backwards causation due to 'ramps' (transition periods).
- ▶ And if we go for a law-preserving rather than a past-preserving treatment of counterfactuals in determinism, there's a worry about long-range backwards causation (of detailed microscopic events).

3. Problem of completeness

- ▶ This is gone.

4. Problem of pre-emption

- ▶ This still arises in a different form. Now the problem is not that Wile E Coyote's lighting of the fuse *does* count as a cause of the explosion, but that Road Runner's lighting it *doesn't*. The explosion would still have happened even if Road Runner hadn't lit the fuse!

Could we get around this by saying that if the bomb had exploded later, it would have been a numerically different explosion? This doesn't seem promising.

5. Problem of chancy causation

- What we say about this depends on what we say about counterfactuals in a chancy universe. If you think 'The bomb wouldn't have exploded if the fuse hadn't been lit' can be true even in a thoroughly indeterministic world where bombs always have some small background chance of going off spontaneously at any time, you're OK.

Refined counterfactual analyses

By far the most effort has been invested in trying to avoid the problem of pre-emption. Lewis's idea was to take the *transitive closure* of counterfactual dependence.

Lewis's analysis

C is a cause of E iff C is disjoint from E and there exist $D_1, \dots, D_n$ disjoint from C , E and each other such that: D_1 counterfactually depends on C , and D_2 counterfactually depends on D_1 , and $\dots$ and E counterfactually depends on D_n .

Chains of dependence

How is this supposed to help with the problem of pre-emption? Lewis was focused on cases of what has come to be called ‘early’ pre-emption, like the following case:

Backup: Boss independently orders Assassin 1 and Assassin 2 to see to it that Victim dies. Both go to Victim’s house. Assassin 1 shoots and kills Victim. Assassin 2, waiting in the wings, sees Assassin 1 shooting, concludes that Victim must be dead, and goes home. If Assassin 1 hadn’t shot at Victim, Assassin 2 would have done so, and Victim would still have died.

Although Victim’s death does not counterfactually depend on Assassin 1’s pulling the trigger, we still have a chain of counterfactual dependence:

- ▶ the death counterfactually depends on Victim’s being hit by Assassin 1’s bullet.
- ▶ Victim’s being hit by Assassin 1’s bullet counterfactually depends on Assassin 1’s pulling the trigger.

Chains of dependence

Unfortunately, this strategy doesn't generalise to other cases of pre-emption, e.g. our example with Wile E Coyote and Road Runner. (These are called cases of 'late pre-emption' in the literature.)

Lewis's strategy also has the controversial consequence that causation is transitive. Many reject this, on the basis of examples like the following

Boulder: A boulder is dislodged and rolls down the mountain towards the trail where I am hiking. I see it coming, duck in time and live to complete the hike.

Clearly, the fall of the boulder is a cause of my ducking. And clearly, my ducking is a cause of my survival. But is the fall of the boulder a cause of my survival?

Holding things fixed

Here's a different thought about the two pre-emption cases.

- ▶ Assassin 1's shooting is a cause of the death because the death counterfactually depends on it *holding fixed the fact that Assassin 2 doesn't shoot*. (I.e.: if A1 hadn't shot *but A2 still didn't shoot*, Victim wouldn't have died.)
- ▶ Road Runner's lighting is a cause of the explosion because the explosion counterfactually depends on it *holding fixed the fact that the last part of the long trail is never lit*. (That is: if RR hadn't lit the short trail and the last part of the long trail had still not been lit, the explosion wouldn't have happened.)

(Hitchcock calls these 'explicitly non-foretracking', or ENF counterfactuals.)

This basic idea has been worked up into an analysis of causation in various ways. The most fruitful implementation, illustrated by Hitchcock (2001), introduces a formal apparatus of 'causal models'.

Worries for simple ENF accounts

Switch: A switch with two positions is wired up to two lights. If the switch is in position A, only light A will be lit; if it is in position B, only light B will be lit. Either way, the room will be illuminated. It's in position A.

If the switch hadn't been in position A but light B still wasn't lit, the room wouldn't have been illuminated. But is the switch's being in position A a cause of the illumination?

Careful Poisoning: (Hitchcock 2007): Assistant Bodyguard puts a harmless antidote in Victim's coffee. Buddy then poisons the coffee, using a poison that is normally lethal, but which is countered by the antidote. Buddy would not have poisoned the coffee if Assistant had not administered the antidote first. Victim drinks the coffee and survives.

If the antidote hadn't been given but the poison still was given, Victim wouldn't have survived. But is the giving of the antidote a cause of the survival?